M Tu W Th F S Su

Moments

formula. $m_{ij} = \sum_{x=1}^{m} \left(\sum_{y=1}^{n} x^i y^j I(x,y) \right)$.

x \ y	1	2	3	4
1	0	0	0	0
2	0	1	1	0
3	0	1	1	0
4	0	0	0	0

compute

$m_{00}, m_{10}, m_{01}, m_{11}$

$$m_{00} = \sum_{x=1}^{m} \left(\sum_{y=1}^{n} I(x,y) \right)$$

Ist = (0+0+0+0) => 0.

IInd = 2.

IIIrd => 2

IVth => 0.

$$\boxed{m_{00} = 4}$$

$$m_{10} = \sum_{x=1}^{m} \left(\sum_{y=1}^{n} x \, I(x,y) \right)$$

Ist => 1 (0) = 0

IInd => 2 (2) = 4.

IIIrd => 3 (2) = 6

IVth => 4 (0) = 0

$$\boxed{m_{10} = 4}$$

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$\cancel{m_{01} = 1(0) = 0}$

$$m_{01} = \sum_{x=1}^{m} \left(\sum_{y=1}^{n} y\, I(x,y) \right)$$

$I^{st} \Rightarrow 1(0) = 0$

$II^{nd} \Rightarrow 2(2) = 4$

$3^{rd} \Rightarrow 3(2) = 6$

$4^{th} \Rightarrow 4(0) = 0$

10

$$\boxed{m_{01} = 10}$$

$$m_{11} \Rightarrow \sum_{x=1}^{m} \left(\sum_{y=1}^{n} xy\, I(x,y) \right)$$

$I^{st} \Rightarrow 1\times1(0) + 1\times2(0) + 1\times3(0) + 1\times4(0) = 0$

$\cancel{4(2) =}$

$II^{nd} \Rightarrow 2\times1(0) + 2\times2(1) + 2\times3(1) + 2\times4 = 0$

$\Rightarrow 4+6 = 10$

$III^{rd} \Rightarrow 3(0) + 6(1) + 9(1) + 12(0)$

$= 15$

$4^{th} \Rightarrow 4(0) + 8(0) + 12(0) + 16(0) = 0$

$$\boxed{m_{11} = 25}$$

$m_{00} \Rightarrow$ area

Centroid $\Leftarrow$

$m_{01} = \frac{m_{01}}{m_{00}} = x_c$

$m_{10} = \frac{m_{10}}{m_{00}} = y_c$